

Mohammad Mahmudul Hasan, PhD

Associate Professor

Department of Electrical and Electronic Engineering



Ibsens gate 13A, Gjøvik 2821, Norway
(+47) 46 34 56 32 / (+880) 1710 037000
mohammad.m.hasan@ntnu.no
NTNU · UITS · Portfolio
Google Scholar · LinkedIn

EDUCATION

- | | |
|--|--------------------------|
|  Norwegian University of Science and Technology, Norway
<i>Doctor of Philosophy</i>
Information and Communication Technology | 05.2022 – 06.2025 |
|  KIIT University, India
<i>Master of Technology</i>
Communication Systems Engineering | 07.2008 – 06.2010 |
|  KIIT University, India
<i>Bachelor of Technology</i>
Electronics and Telecommunication Engineering | 07.2004 – 06.2008 |

WORK EXPERIENCE

- | | |
|--|---|
|  University of Information Technology and Science, Bangladesh
✓ Associate Professor, Electrical and Electronic Engineering
✓ Assistant Professor, Electrical and Electronic Engineering
✓ Advisor, Information and Communication Technology Cell
✓ Head of the Department, Electronics and Communication Engineering
✓ Director, Information and Communication Technology Cell
✓ Project Manager, Enterprise Resource Planning
✓ Editorial Board Member, Journal of Science and Engineering
✓ Member, Self-Assessment Committee, IQAC
✓ Member, Finance Committee | 01.2011 – Present
07.2025 – Present
01.2011 – 05.2022
01.2018 – Present
02.2019 – 04.2022
12.2020 – 04.2022
03.2020 – 04.2022
10.2020 – 04.2022
07.2017 – 07.2018
03.2020 – 04.2022 |
|  Brno University of Technology, Czech Republic
✓ Visiting Research Fellow, Department of Automation | 08.2024 – 01.2025 |
|  Norwegian University of Science and Technology, Norway
✓ Research Fellow, Faculty of Engineering | 05.2022 – 06.2025 |
|  Sanyo Engineering & Construction Inc., Japan
✓ Industrial Trainer, BJIT Limited, Bangladesh | 04.2018 – 01.2019 |
|  KIIT University, India
✓ Assistant Professor, Electronics and Telecommunication Engineering
✓ Teaching Assistant, Electronics and Telecommunication Engineering | 02.2009 – 12.2010
07.2010 – 12.2010
02.2009 – 06.2010 |

SELECTED PUBLICATIONS

- | | |
|---|---------------|
|  Inaamullah Khan, Mohammad Mahmudul Hasan , Michael Cheffena Gebresilassie, “Transmitter-Assisted Joint Data-Aided Channel Estimation and PAPR Reduction Scheme in Wireless Fading Channels”, Scientific Reports , vol. 16, no. 1, 8015, 2026 | (2026) |
|  Mohammad Mahmudul Hasan , Onur Alev, Pavel Skrabanek, Gabriela Soukupová, Fatima Hassouna, and Michael Cheffena Gebresilassie, “Microwave MIMO E-Nose for Wireless Communication and Selective Detection of VOC Mixtures with Concentration Estimation”, ACS Sensors , vol. 10, no. 9, pp. 6446–6463, 2025 | (2025) |
|  Inaamullah Khan, Mohammad Mahmudul Hasan , Michael Cheffena Gebresilassie, “A Novel Low-Complexity Peak-Power-Assisted Data-Aided Channel Estimation Scheme for MIMO-OFDM Wireless Systems”, IEEE Open Journal of Signal Processing , vol. 6, pp. 992 – 1003, 2025 | (2025) |
|  Onur Alev, Mohammad Mahmudul Hasan , Emel Tuğba Ertuğrul, Selçuk Birdoğan, Okan Özdemir, Eda Goldenberg, Michael Cheffena, “Hydrothermally Synthesized Molybdenum disulfide Nanoflakes: Structural, Electrical, and Antenna-based Gas Sensing Characteristics”, Sensors & Actuators: A. Physical , vol. 393, 116756, 2025 | (2025) |

1. **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, and Michael Cheffena Gebresilassie, "Molecularly Imprinted Polymer-Based Electronic Nose for Ultrasensitive, Selective Detection and Concentration Estimation of VOC Mixtures", **IEEE Sensors Journal**, vol. 25, no. 10, 2025 (2025)
2. **Mohammad Mahmudul Hasan**, Todd Cowen, Onur Alev and Michael Cheffena Gebresilassie, "MIMO Microwave Sensor for Selective and Simultaneous Detection of Methanol and Ethanol Gases at Room Temperature," **IEEE Transactions on Instrumentation & Measurement**, vol. 74, 9511613, 2025 (2025)
3. **Mohammad Mahmudul Hasan**, Onur Alev and Michael Cheffena Gebresilassie, "Dual-Functional Antenna Sensor for Highly Sensitive and Selective Detection of Isopropanol Gas Using Optimized Molecularly Imprinted Polymers," **ACS Sensors**, vol. 10, no. 3, pp. 2147–2161, 2025 (2025)
4. **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena Gebresilassie, "MoS₂/MoO_x Nanoflake-Based Dual-Functional Antenna Sensors for Highly Sensitive and Selective Detection of Volatile Organic Compounds," **ACS Applied Nano Materials**, vol. 7, no. 21, pp. 25065–25077, 2024 (2024)
5. **Mohammad Mahmudul Hasan**, Todd Cowen and Michael Cheffena Gebresilassie, "A Novel Molecularly Imprinted Polymer-Based Carbon Nanotube-Coated Microwave Sensor for Selective Detection of Methanol Gas," **IEEE Sensors Letters**, vol. 8, no. 5, 6004904, 2024 (2024)
6. **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena Gebresilassie, "A Novel Molybdenum Disulfide-Based High-Precision Microwave Sensor for Methanol Gas Detection at Room Temperature," **IEEE Microwave and Wireless Technology Letters**, vol. 34, no. 6, pp. 691 – 694, 2024 (2024)
7. **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, "Adaptive Antenna Impedance Matching Using Low-Complexity Shallow Learning Model", **IEEE Access**, vol. 11, pp. 74101 – 74111, 2023 (2023)
8. **Mohammad Mahmudul Hasan**, Michael Cheffena Gebresilassie, Slobodan Petrovic, "Physical-layer Security Improvement in MIMO OFDM Systems using Multilevel Chaotic Encryption", **IEEE Access**, vol. 11, pp. 64468 – 64475, 2023 (2023)
9. Inaamullah Khan, Michael Cheffena Gebresilassie, **Mohammad Mahmudul Hasan**, "Data Aided Channel Estimation for MIMO-OFDM Wireless Systems Using Reliable Carriers", **IEEE Access**, vol. 11, pp. 47836 – 47847, 2023 (2023)
10. **Mohammad Mahmudul Hasan**, Mohammad Mahadi Hasan Foad, "Modified Gamma Correction Companding for PAPR Reduction in OFDM Systems Considering Solid State Power Amplifier and Wireless Channels", **Circuits, Systems, and Signal Processing**, vol. 37, no. 10, pp. 4431- 4454, 2018 (2018)
11. **Mohammad Mahmudul Hasan**, Mohammad Mahdi Hasan Faisal, "IGCC for PAPR Reduction in OFDM Systems over the Nonlinearity of SSPA and Wireless Fading Channels", **Circuits, Systems, and Signal Processing**, vol. 35, no. 8, pp. 2855–2880, 2015 (2015)
12. **Mohammad Mahmudul Hasan**, "PAR Reduction in SU/MU-MIMO OFDM Systems using OPF Precoding over the Nonlinearity of SSPA", **Wireless Personal Communications**, vol. 83, no. 3, pp. 2225-2248, 2015 (2015)
13. **Mohammad Mahmudul Hasan**, "A Novel CVM Precoding Scheme for PAPR Reduction in OFDM Transmissions", **Wireless Network**, vol. 20, no. 6, pp. 1573-1581, 2014 (2014)
14. **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Scheme for OFDM Systems Based on Gamma Correction", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1655-1668, 2014 (2014)
15. **Mohammad Mahmudul Hasan**, "A New PAPR Reduction Technique in OFDM Systems Using Linear Predictive Coding", **Wireless Personal Communications**, vol. 75, no. 1, pp. 707-721, 2014 (2014)
16. **Mohammad Mahmudul Hasan**, "VLM Precoded SLM Technique for PAPR Reduction in OFDM Systems", **Wireless Personal Communications**, vol. 73, no. 3, pp. 791-801, 2013 (2013)
17. **Mohammad Mahmudul Hasan**, "PAPR Reduction in OFDM Systems Based on Autoregressive Filtering", **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1637-1654, 2013 (2013)

PROFESSIONAL CERTIFICATIONS

📄 Brno University of Technology, Czech Republic Professional Competence, Electrical Engineering – NV No. 194/2022 Coll.	08.2024 – 01.2025
📄 GrameenPhone Ltd., Bangladesh Intern Engineer, Transmission Planning Division	07.2007 – 10.2007
📄 All India Radio & Doordarshan (Prasar Bharati), India Industrial Training, Broadcasting Corporation of India	05.2007 – 07.2007
📄 Red Hat Bhubaneswar, India Industrial Training, Red Hat Linux RHEL 4	04.2006 – 07.2006

GRANTS & AWARDS

🏆 Research Recognition Nominated for <i>Best Doctoral Thesis</i> in Sensors Norwegian University of Science and Technology	06.2025
🏆 Research Grant Awarded 146 000 NOK The Research Council of Norway	08.2024
🏆 Chancellor's Gold Medal Awarded for <i>securing the highest CGPA (10/10)</i> KIIT University, India	12.2010
🏆 Founder's Gold Medal Awarded for securing the <i>first position</i> in Master of Technology KIIT University, India	12.2010
🏆 Chancellor's Merit Scholarship Awarded for <i>academic excellence</i> , Bachelor's and Master's studies KIIT University, India.	2004 – 2010

LANGUAGE PROFICIENCY

☑ Bengali (native), English (fluent), Hindi (fluent), Arabic (A1), and Norsk (A2/B1)

CITIZENSHIP & RESIDENCY

📄 Bangladeshi (Citizen), Norway (Permanent Residency)

REFERENCES

👤 **Dr. Michael Cheffena Gebresilassie**
(PhD Supervisor)
Professor, Faculty of Engineering,
NTNU, Gjøvik 2815, Norway
✉ michael.cheffena@ntnu.no, 📞 (+47) 45226765

👤 **Dr. Alok Mishra**
(Distinguished Professor)
Professor, Faculty of Engineering,
NTNU, Gjøvik 2815, Norway
✉ alok.mishra@ntnu.no, 📞 (+47) 46665761

I certify that the above statements are true to the best of my knowledge.

– *Mohammad Mahmudul Hasan*